

Joint modeling of direct and social genetic effects in mono-genotypic and pluri-specific groups: case study in intercrops

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1 Simulations of field trials

1.1 Default parameter values

- Mean yield of wheat genotypes (in q/ha): $\mu^{(f)} = 65$ in sole crops, and reduced by half in intercrops (65×0.5)
- Mean yield of pea genotypes (in qt/ha): $\mu^{(t)} = 30$
- Wheat DBV variance: $\sigma_{D(f)}^2 = (CV_g^{(f)} \times \mu^{(f)})^2$ where $CV_g^{(f)} \in \{8\%, 8.5\%\}$
- Prop. of wheat SBV variance: $\sigma_{S(f)}^2 = 0.2 \sigma_{D(f)}^2$
- Prop. of wheat SIGV variance: $\sigma_{SIGV(f)}^2 = 0.5 \sigma_{D(f)}^2$
- Correlation between wheat DBV and SBV: $\rho_{DS(f)} = -0.8$
 - Covariance: $\sigma_{DS(f)} = \rho_{DS(f)} / (\sigma_{D(f)} \times \sigma_{S(f)})$
- Variance of DBVxSBV: $\sigma_{DS}^2 = 0$
- Variance of BV^{SC} : $\sigma_{BV^{SC}}^2 = \sigma_{D(f)}^2 + \sigma_{SIGV(f)}^2$
- Variance of BV^{IC} : $\sigma_{BV^{IC}}^2 = \sigma_{D(f)}^2 + 2 \sigma_{DS(f)} + \sigma_{S(f)}^2$
- Sole error variance for focal: $\sigma_{e,sc(f)}^2 = \frac{1-h_{sc(f)}^2}{h_{sc(f)}^2} \times (\sigma_{BV(SC,f)}^2 \times r)$ where $h_{sole(f)}^2 = 0.7$, and r , the number of replicates (2)
- Intercrop error variance was approxiamte for both, tester and focal by:
 $\sigma_{e,ic}^2 = \frac{1-h_{ic}^2}{h_{ic}^2} \times (\sigma_{BV(IC)}^2 \times (t \times r))$ where $h_{ic}^2 = 0.7$, t the numer of effective tester, and r the number of replicates
- Correlation between mixed focal and tester errors: $\rho_{e,ic(f,t)} = -0.2$